

IPBMA - Practical case 2: Methodology for analyzing a cardiovascular optical coherence tomography (OCT)

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I. INTRODUCTION

The goal of this practice is to implement a methodology to analyze cardiovascular optical coherence tomography (OCT) imaging for identifying the aorta's wall and finding struts within it. The image set provided consists of 15 images in "png" format their dimensions are the same in all of them (704x704). Images are snapshot taken during different moments of the camera's path while going through the artery.

II. METHODOLOGY

This part is going to be explained the process and the chosen techniques for getting the parameters of the aorta in each image, the detection of struts in the wall and the struts categorization by size. The methodology was divided into 5 tasks that are explored in the following sections:

A. Pre-processing of the image

The first step was to extract the region of interest within the image. After several cropping trials, finally, it was decided to select the area within the next ranges: y-axis between 125 and 450, and x-axis between 175 and 550. An example of the cropping is shown in Figure 1.

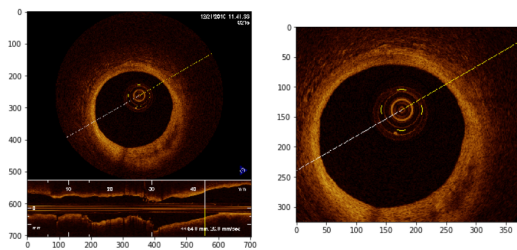


Fig. 1. Cropping of region of interest

Once, the region of interest was gotten, next steps were to eliminate superfluous structures and noise from the image. The steps followed were applied: a Contrast Limited Adaptive Histogram Equalization, thresholding segmentation, an erosion of the image and a Canny edge detector; an example of the

process it is shown in Figure 2. Below, the techniques and the parameters used are explained in detail.

1) *Contrast Limited Adaptive Histogram Equalization (CLAHE)*: For obtaining an image with higher global contrast, a histogram equalization (HE) is recommended. Actually, for further histogram equalization improvements, an algorithm called Contrast Limited Adaptive Histogram Equalization (CLAHE) it might be applied. Contrast Limited Adaptive Histogram Equalization (CLAHE) is a variant of Adaptive Histogram Equalization, where the adaptive method computes several histograms, each corresponding to a distinct section of the image, and uses them to redistribute the lightness values of the image. However, AHE has a tendency to overamplify noise in relatively homogeneous regions of an image, CLAHE prevents this by limiting the amplification. So, a CLAHE was applied where the parameters used for the method were: 10.0 as a threshold for contrast limiting and (8,8) as the size of the tiles for applying the histogram equalization.

2) *Thresholding*: After the equalization of the image, it was sought to convert the grayscale image to a binary image, in order to segment the foreground, the aorta walls in this case, and the background of the input image. For that, a global thresholding technique was used for this segmentation using a threshold value of 11 and a Binary method.

3) *Erosion*: In order to remove some noise such as loose points or the central circles generated by the camera. A technique from morphological operations was used, specifically an Erosion, applying a structuring element of 13x13.

4) *Canny*: As last step in the preprocessing part, it was applied a Canny algorithm to find edges, mainly interested in the circular boundaries. Basically, it does edge detection by using linear filtering with a Gaussian kernel to reduce noise in order to compute the strength and direction of the edges for each pixel in the smoothed image. For the templates, using a sigma with value 3.

B. Location of the aortic wall

For location the aortic wall, as first step a Hough transformation was used for isolating circles within the snapshot. The

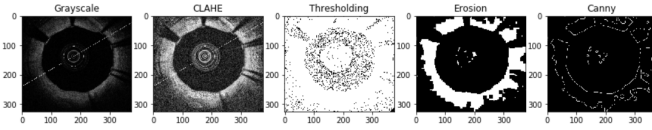


Fig. 2. Steps for image pre-processing

Hough method works as a feature extraction technique that converts an input image from Cartesian to polar coordinates. With this, the center and the radius of a circle gotten after the Hough application, were the starting parameters to start calculating the real shape of the walls of the aorta in the image. A function was created with the goal of searching the first point found where there was a border, taking as parameters: center, radius and a defined number of angles, 180 grades for our case. For that, through a loop, given the center calculated with Hough, an angle, starting by 0, and a radius between two-thirds and twice the input radius, a point is gotten. Then, it is verified whether the value is 255, if so, the point is saved and the border point for the next angle is calculated. In the case that none border was found, the point would take the same radius that the previous angle. A very specific issue was found when executing, when the first angle (0 degrees) can not found a border, then, the input Hough radius is assigned.

An example of the wall detection result is shown in Figure 3.

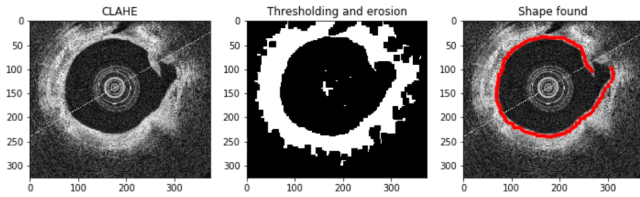


Fig. 3. Aortic wall detection

C. Aortic parameters

The process before was applied for all the images, with it, some aortic parameters were gotten as it is shown in Figure 4: radius, area, appearance and its categorization (irregular or non-irregular), degree of circularity.

D. Struts' identification

For identification the struts present in the aorta's wall the region of interest was augmented, with this the area would be within the next ranges: y-axis between 100 and 450, and x-axis between 175 and 550. It was created a function that was analyzing several circumferences of the image given, specifically: points of the area detected above, points of the circumference found with the Hough transformation, as well as same circumference but augmented the radius in 5 points, 10 points and 15 points. The idea was to go analyzing candidates by different degrees of depth in relation to the radius. Before searching the candidates, the points that make

	file	radius	area	appearance	appearance cat	degree_circularity
0	m001-IMG002-00038.png	95.800000	28832.406401	9.035977	irregular	0.989189
1	m001-IMG002-00042.png	93.516667	27474.380546	5.165564	non-irregular	0.973684
2	m001-IMG002-00074.png	105.772222	35147.394032	8.556239	irregular	1.111111
3	m001-IMG002-00075.png	108.755556	37158.038055	9.978099	irregular	1.164251
4	m001-IMG002-00076.png	111.522222	39072.635156	10.077177	irregular	1.153110
5	m001-IMG002-00077.png	114.277778	41027.348068	7.816688	irregular	1.118721
6	m001-IMG002-00082.png	121.155556	46114.397570	9.485558	irregular	1.081897
7	m001-IMG002-00083.png	121.522222	46393.942302	9.794247	irregular	1.086580
8	m001-IMG002-00088.png	123.300000	47761.287537	5.096731	non-irregular	1.047009
9	m001-IMG002-00089.png	124.288889	48530.468489	6.642885	irregular	1.067511
10	m001-IMG002-00091.png	125.400000	49402.047133	6.045751	irregular	1.054167
11	m001-IMG002-00093.png	124.111111	48391.735758	4.515289	non-irregular	1.028926
12	m001-IMG002-00198.png	120.372222	45520.018421	2.736645	non-irregular	1.029661
13	m001-IMG002-00199.png	119.616667	44950.369327	3.055551	non-irregular	1.029787
14	m001-IMG002-00215.png	125.911111	49805.578115	4.634759	non-irregular	1.024291

Fig. 4. Aortic parameters for each image

up each of the circles were listed either as 0 or 255, where the 0 value would be considered as possible strut found. Thus, in the 1st degree of possible candidates, composed by the shape area and the Hough circumference, the points detected were included directly. In the 2nd degree, consisted by points from circumference with radius + 5 and circumference with radius + 10, points that were in the 1st degree and next to it, below or above, were selected as candidates. Finally, as 3rd degree, points from circumference with radius + 20, the same rule as before is applied, points that were in the 2nd degree and next to it, below or above, were selected as candidates.

After the selection of possible candidates, a filtering was done in order to clean and get a final detection of struts. Basically, this process for selecting the candidates was a simple threshold for limiting the minimum and maximum number of points to be considered as strut. The thresholds to be selected were: to have more than 1 point in the third degree and less than 5 in the first degree.

In Figure 5 the process explained above is shown. In the second image, the possible candidates for each degree are drawn: blue for 1st degree, red for second degree and yellow for candidates in the third degree. Last image, it is showing the final results after the filtering done.

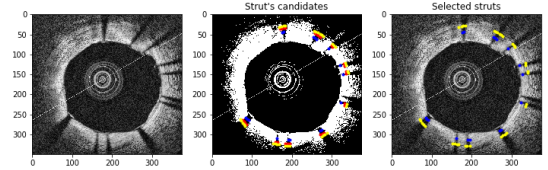


Fig. 5. Process for strut's identification

However, this method has some limitations, in the case of Figure 6, it is able to see the limitation of not considering candidates composed with only one point. This means, that, currently, taking into account only those candidates that are composed for more than one point in the third degree, some

struts are being discarded.

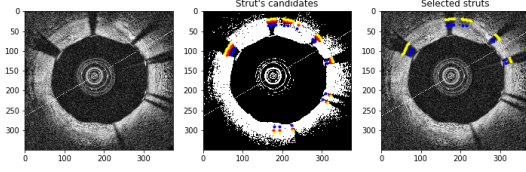


Fig. 6. Limitations for lower limit filtering

Regarding the upper limit, Figure 7 shown one of the cases that are being filtered correctly.

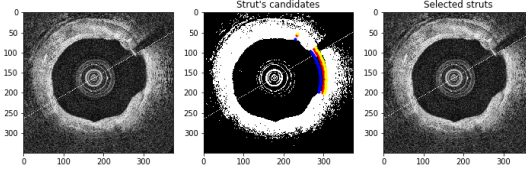


Fig. 7. Upper limit filtering

E. Struts' categorization

Finally, for the categorization of the detected struts according to their size, when a strut was composed for more than 4 points in the first degree, it was classify as big otherwise as small. In Table 8 the total number of struts, the number of small and big ones and their categories are shown for each image.

file	total	big struts	small struts	struts category
0 m001-IMG002-00038.png	0	0	0	{}
1 m001-IMG002-00042.png	5	1	4	{2: 'small', 5: 'big', 6: 'small', 7: 'small', 8: 'small'}
2 m001-IMG002-00074.png	1	1	0	{3: 'big'}
3 m001-IMG002-00075.png	2	0	2	{4: 'small', 5: 'small'}
4 m001-IMG002-00076.png	0	0	0	{}
5 m001-IMG002-00077.png	0	0	0	{}
6 m001-IMG002-00082.png	5	0	5	{3: 'small', 5: 'small', 7: 'small', 8: 'small', 9: 'small'}
7 m001-IMG002-00083.png	5	2	3	{6: 'big', 7: 'big', 8: 'small', 10: 'small', 11: 'small'}
8 m001-IMG002-00088.png	3	0	3	{1: 'small', 4: 'small', 5: 'small'}
9 m001-IMG002-00089.png	10	0	10	{1: 'small', 2: 'small', 3: 'small', 4: 'small', 5: 'small', 6: 'small', 7: 'small', 8: 'small', 9: 'small', 10: 'small'}
10 m001-IMG002-00091.png	9	0	9	{1: 'small', 2: 'small', 3: 'small', 4: 'small', 5: 'small', 6: 'small', 8: 'small', 10: 'small', 11: 'small'}
11 m001-IMG002-00093.png	4	0	4	{1: 'small', 2: 'small', 3: 'small', 4: 'small'}
12 m001-IMG002-00198.png	0	0	0	{}
13 m001-IMG002-00199.png	1	0	1	{1: 'small'}
14 m001-IMG002-00215.png	0	0	0	{}

Fig. 8. Categorization detected struts according the size

III. DISCARDED TECHNIQUES

1) *Lower and upper limit for selecting struts:* Figure 9 and Figure 10 are showing some examples of changing the threshold for considering a strut candidate as valid or not.

In the first Figure, the lower limit would be zero, which means that any candidate with one or points in the third degree, is being accepted, this was the reason of why this limit was discarded.

In the second Figure, upper limit is 6, meaning that any candidate with more than 6 point in the first degree is accepted.

The issue with this was that due to the irregularity in some of the images, some of these openings were being accepted and for avoiding this, this limit was discarded.

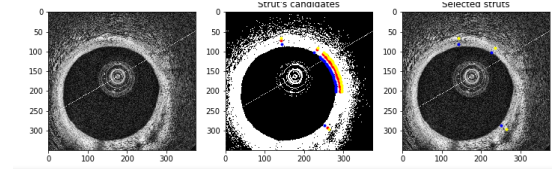


Fig. 9. Minimum number of points for selecting a strut higher or equal to 1

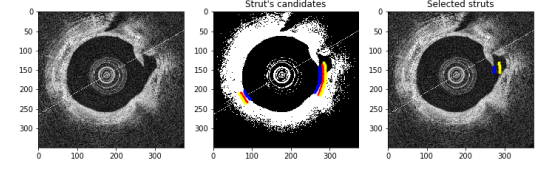


Fig. 10. Maximum number of points for selecting a strut higher than 6